

Mohan Ram Vemuri

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PROFESSIONAL SUMMARY:

Generative AI Engineer with 6+ years of experience driving the development and deployment of intelligent, scalable, and data-driven solutions. Proven expertise in designing and optimizing large language model (LLM) pipelines, RAG frameworks, and agentic AI systems, consistently delivering measurable improvements in efficiency, accuracy, and user engagement across diverse industries. Expert in Python, cloud platforms (AWS, Azure, GCP), and MLOps practices.

SKILLS:

GenAI & AI Frameworks: OpenAI, Gemini, LLaMA, Claude, GPT-4+, Anthropic, CrewAI, LangGraph, LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLMOps, Vector Databases (Chroma DB, FAISS, Weaviate, Pinecone), n8n automation, A/B testing, LoRA, QLoRA, Model Context Protocol (MCP)

ML Frameworks: Scikit-learn, TensorFlow, PyTorch, Keras, MLFlow, NumPy, Pandas, Natural Language Processing (NLP),

Medical terminology: HL7, FHIR, ICD, EHR, HIPAA

Programming Languages: Python 3, Java, JavaScript, React JS, Node JS, Type Script, Perl, Linux, PowerShell scripting

Tools & Platforms: Docker, Jenkins, Kubernetes, REST APIs, Fast API, Kafka, Flask

Cloud: Azure (AI Foundry, Fabric-Data Factory, Delta Lake, Power BI, Synapse Analytics, Machine Learning Studio, Copilot), AWS (S3, Lambda, SageMaker, RedShift, API Gateway), GCP (Big Query, Firebase, GCS), Databricks, Snowflake, Terraform

Databases: MySQL, Postgres, NoSQL, MongoDB, Redis.

Certifications: Oracle Gen AI Professional, Kaggle Generative AI Certificate, Snowflake platform Fundamentals, NVIDIA Deep Learning Fundamentals, Scaled Agile Framework for Enterprise (**SAFe-5**) Agile Practitioner, Udemy MS Azure Fabric

WORK EXPERIENCE:

ASTORIA AI (Startup) – GENERATIVE AI ENGINEER

AUG 2025 – Present

- Led 0→1 development of a modular, multi-provider AI agent platform in Python, integrating OpenAI and Groq LLMs with dynamic selection, achieving a 20% cost reduction and 15% latency improvement for career guidance.
- Designed and implemented specialized AI agents (JobAdvisor, ResumeParser, SkillAssessment, CareerPath) within an extensible FAST API framework, automating end-to-end career guidance processes.
- Engineered an automated ETL pipeline using Docker, Kestra, and JSearch API for real-time job extraction and caching with Azure and Postgres, enhancing job recommendation relevance by 30%.
- Worked on end to end deployments of AI pipelines using Azure Container Instances(ACI), Azure Container Apps (ACA), Logic Apps and Azure Container Registry with Github Actions.
- Created CareerPath Agent to recommend actionable career trajectories based on skill graph, assessments and goals.
- Developed a high-accuracy ResumeParser agent utilizing Pydantic schemas for strict data validation, achieving 95% accuracy in structured JSON output for diverse resume formats.
- Built and deployed a WebSocket-based real-time update system with a React front end and NodeJS backend, providing instant parsing status notifications to users, improving engagement by 25%.

UNIVERSITY OF HOUSTON – GRADUATE ASSISTANT

SEPT 2024 – MAR 2025

- Developed an LLM-powered biomedical search engine for a pharmaceutical department, using EHR and HL7 standards to improve data accuracy. This reduced researchers' literature discovery time by 95% through efficient retrieval of relevant medical publications and patient data.
- Built a modular, agent-based system using LangGraph that significantly improved search accuracy across PubMed by automating the extraction and validation of medical information, including ICD codes and other healthcare-specific data formats. This system also used the BIOBERT transformer and restructured queries for better relevance.
- Integrated a Retrieval-Augmented Generation (RAG) pipeline with Gemini 2.0-Flash LLM, fetching and filtering real-time metadata from the PubMed API. The pipeline's structured JSON schema ensures data precision and is compliant with FHIR standards for enhanced interoperability.

FIDELITY INFORMATION SERVICES (FIS GLOBAL) - LEAD ENGINEER**AUG 2021 – AUG 2023**

- Engineered an automated ServiceNow bot with NLP hooks, classifying IT Service Management (ITSM) events and generating tickets post-incident, resulting in an 80% reduction in manual triage efforts.
- Designed and implemented an end-to-end monitoring system for 1,000+ distributed systems, tracking OS internals, CPU, memory, I/O, and networking metrics, improving proactive anomaly detection by 40% and reducing critical incident response time by 25%.
- Containerized monitoring agents with Docker and implemented CI/CD workflows using Jenkins and OpenShift, ensuring reliable, version-controlled production rollouts.
- Leveraged the Decisions RPA platform to automate rule-based validation across IT and financial domains, reducing manual effort by 60% and human error by 35% in critical processes.
- Automated ingestion and preprocessing of Excel files via AWS Lambda triggers, applying rule-based checks and color-coded error highlighting to ensure high data integrity.
- Managed weekly batch jobs for mainframe updates and developed Python scripts to parse logs, verify success, and automate stakeholder notifications, ensuring 99.9% data consistency.

WIPRO LTD, INDIA- SENIOR PROJECT ENGINEER**MAY 2018 – AUG 2021**

- Engineered high-throughput test automation pipelines (processing 100GB/s with 1M+ IOPS) using Python and Perl (OOP), integrating with ONTAP OS at the kernel level, accelerating testing cycles by 50%.
- Migrated legacy frameworks to Python, enabling system-level introspection and real-time metric collection from diverse storage protocols (SAS, NVMe-oF, ZNS), improving data visibility by 70%.
- Developed regression analysis pipelines that logged performance trends and auto-published test summaries to Confluence, enabling early detection of performance degradation and preventing an average of 3 critical bugs per release cycle.
- Built batch automation infrastructure for remote drive qualification with cron-jobs, successfully simulating complex distributed environments for 1000+ devices.

PROJECTS:

Full-Stack: Personalized AI Interview Agent_(Groq-Llama 3, Mistral, Fast API, React, Python, Voice AI, Tailwind, MongoDB, Prompt Engineering)

- Designed and implemented a personalized AI interview agent using Groq to simulate adaptive, real-time interviews based on candidate resumes and job descriptions.
- Developed modular Python services for resume parsing, JD analysis, question generation, and behavioral feedback using Llama 3 models, integrated with MongoDB for persistent storage.
- Built a Fast API backend with session memory, scoring engine, and feedback loops; integrated a React & Java Script based frontend for interactive, voice-enabled candidate interviews.
- Engineered tone detection and coaching feedback modules to analyze emotional sentiment in candidate responses and dynamically adjust interview style in real-time.

MLOps: Youtube sentiment analysis Pipeline_(MLflow, DVC, Docker, AWS, AI-ML, Chrome Plugin)

- Built and deployed an end-to-end sentiment analysis pipeline using DVC (dataset version control) and ML-flow for experiment tracking and model governance with Machine Learning and artificial intelligence
- Containerized the inference service with Docker and deployed it on AWS EC2, ensuring scalable access via REST API and integrated the API with a Chrome Extension, enabling real-time predictions.
- Implemented full CI/CD automation with GitHub Actions, ensuring continuous testing, training, and deployment.

EDUCATION

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- **UNIVERSITY OF HOUSTON, USA** **May 2025**
Master of Science (MS) in Engineering Data Science (*Dean's Merit Scholarship awardee**)
 - **K L UNIVERSITY, INDIA- Bachelors in Electronics and Computer Engineering** **May 2018**